

Numerical skills for neuroscience

Part 2 · Vectors, matrices and PCA

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From many measurements to a simpler picture

Tool	What it lets us do
Vector	Collect measurements into one object
Matrix	Organise data or combine measurements
Projection	Describe an observation along a chosen direction
PCA	Find directions that capture the most variation

Keep asking: what do the **rows**, **columns** and **axes** represent?

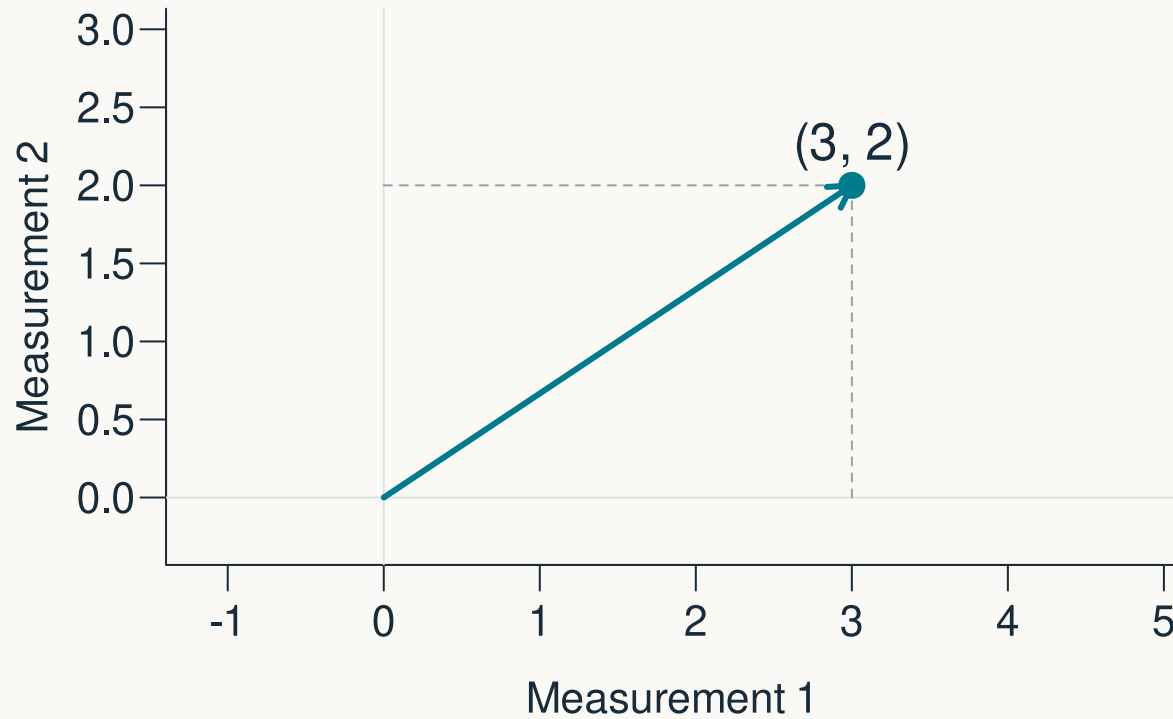
A data matrix has rows and columns

	Neuron 1	Neuron 2	Neuron 3
Trial 1	4	8	2
Trial 2	6	9	4
Trial 3	3	6	1
Trial 4	7	10	5

← One row:
one observation

$\mathbf{X} \in \mathbb{R}^{n \times p}$ n observations, p variables

A vector is a list and a point



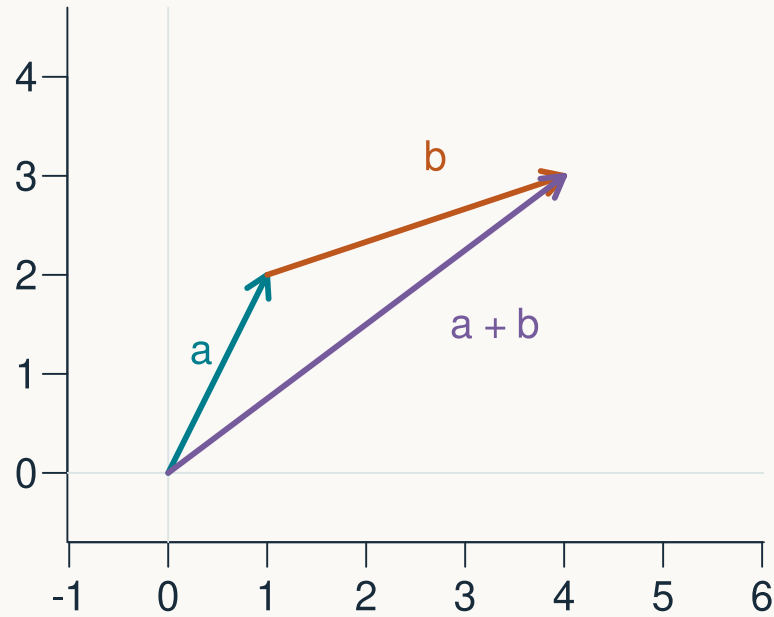
$$\mathbf{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Two measurements → one point in **two dimensions**.

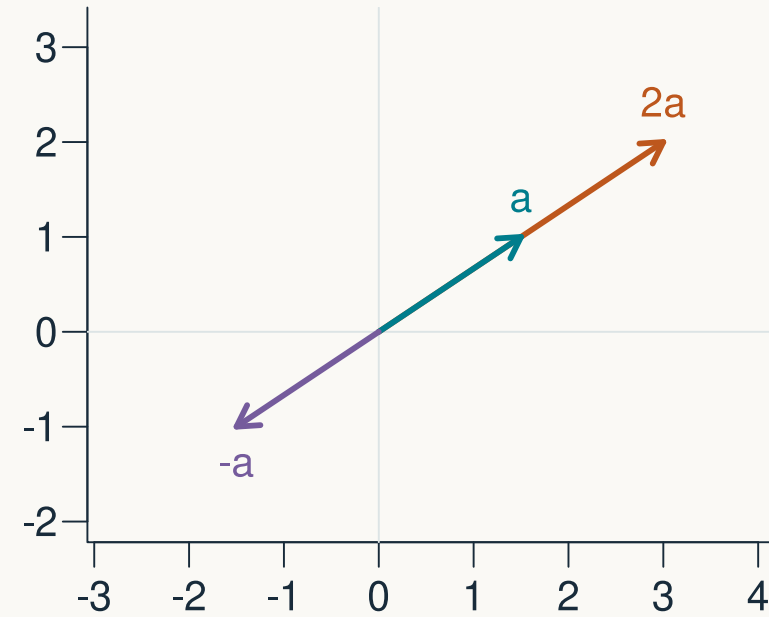
An arrow from the origin gives the same coordinates.

Add vectors; multiply them by a scalar

Add: head to tail



Scale: length and possibly direction



$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$2 \begin{bmatrix} 1.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

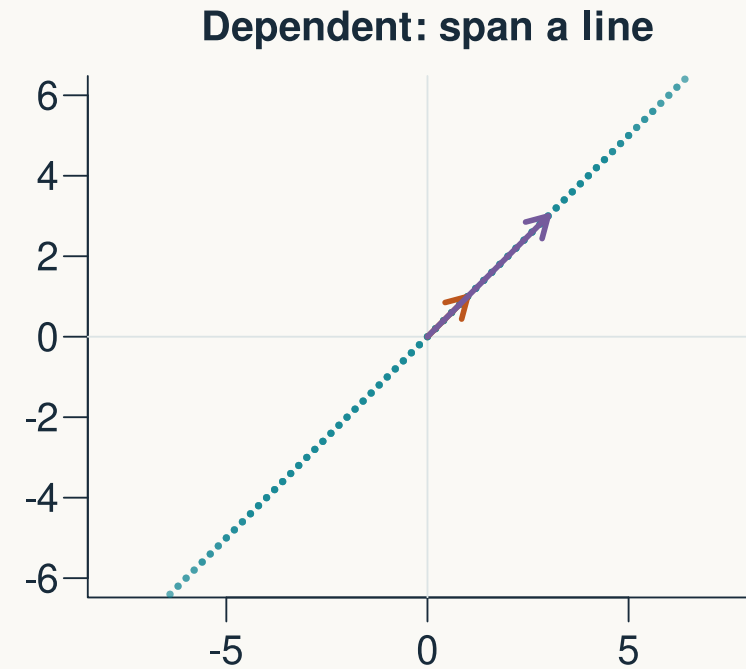
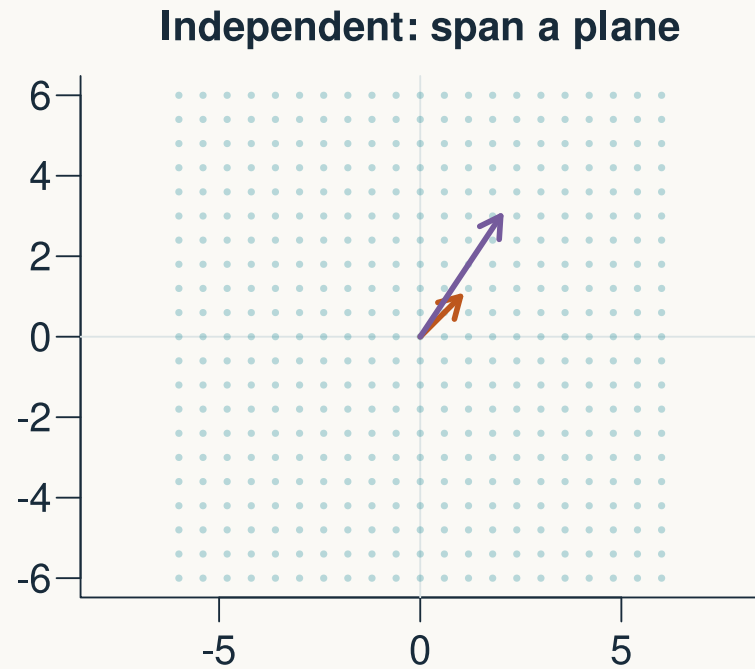
A linear combination is a weighted sum

$$c\mathbf{v} + d\mathbf{w} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} c + 2d \\ c + 3d \end{bmatrix}$$

Set $c = 2$ and $d = -1$. Where do you land?

$$2\mathbf{v} - \mathbf{w} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

Span: everywhere those combinations can reach



A **basis** is an independent set that spans the space being described.

A matrix packages the columns we want to combine

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = [\mathbf{v} \quad \mathbf{w}]$$

$$\underbrace{\mathbf{A}}_{2 \times 2} \underbrace{\begin{bmatrix} c \\ d \end{bmatrix}}_{2 \times 1} = c\mathbf{v} + d\mathbf{w}$$

Matrix $\times$ vector = a weighted sum of the matrix's **columns**.

A dot product multiplies, then adds

$$\mathbf{a}^T \mathbf{b} = a_1 b_1 + a_2 b_2 + \cdots + a_p b_p$$

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 1 \times 3 + 2 \times 1 = 5$$

The result is **one number**.

Dot products also describe alignment



$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

In 2D: $\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2}$. The angle requires **nonzero** vectors.

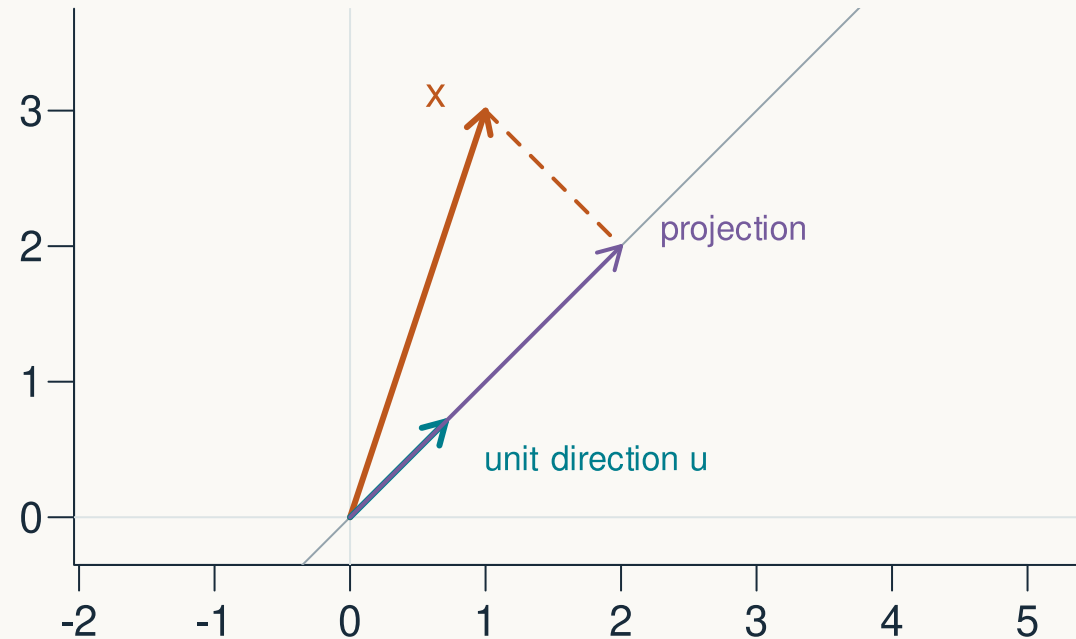
Projection gives a coordinate along an axis

For a **unit** direction $\mathbf{u}$:

$$z = \mathbf{x}^T \mathbf{u}$$

$$\text{projected vector} = z\mathbf{u}$$

One number locates the point along this line.



Transpose swaps rows and columns

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \implies \mathbf{A}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$2 \times 3 \longrightarrow 3 \times 2$$

In R: `t(A)`.

Matrix multiplication: a row meets a column

A			B			AB	
1	1	x	2	2	=	5	6
2	-1		3	4		1	0

Row 1 of A . column 2 of B: $1 \times 2 + 1 \times 4 = 6$

The same product, one column at a time

$$\mathbf{AB} = [\mathbf{Ab}_1 \quad \mathbf{Ab}_2]$$

$$\mathbf{A} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

Each column of the product is a combination of columns of $\mathbf{A}$.

Check the dimensions before multiplying

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{B}}_{n \times p} = \underbrace{\mathbf{C}}_{m \times p}$$

$(100 \times 3)(3 \times 2) \rightarrow$ what shape?

Can we reverse the order?

100×2 . Reversed: inner dimensions 2 and 100 do not match.

In R, `*` and `%*%` do different jobs

Element by element

```
1 A <- matrix(c(1, 1, 2, -1), 2, byrow = TRUE)
2 B <- matrix(c(2, 2, 3, 4), 2, byrow = TRUE)
3 A * B
```

	[,1]	[,2]
[1,]	2	2
[2,]	6	-4

Matrix multiplication

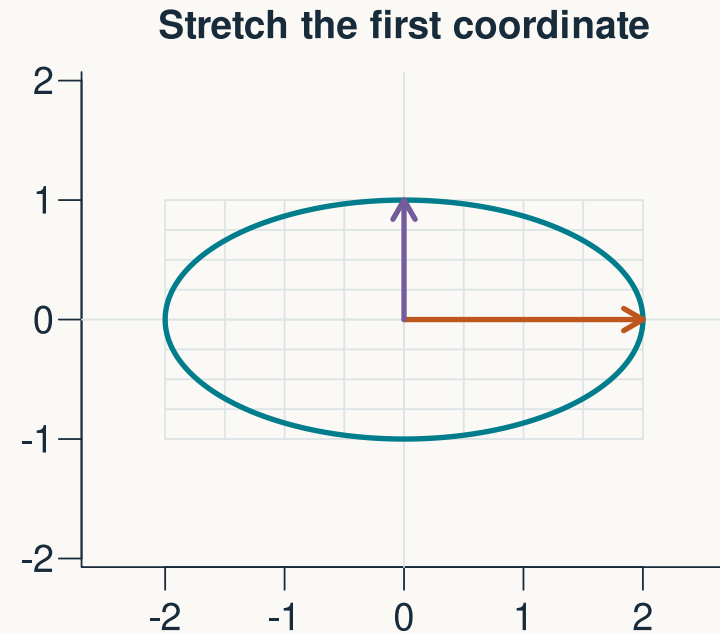
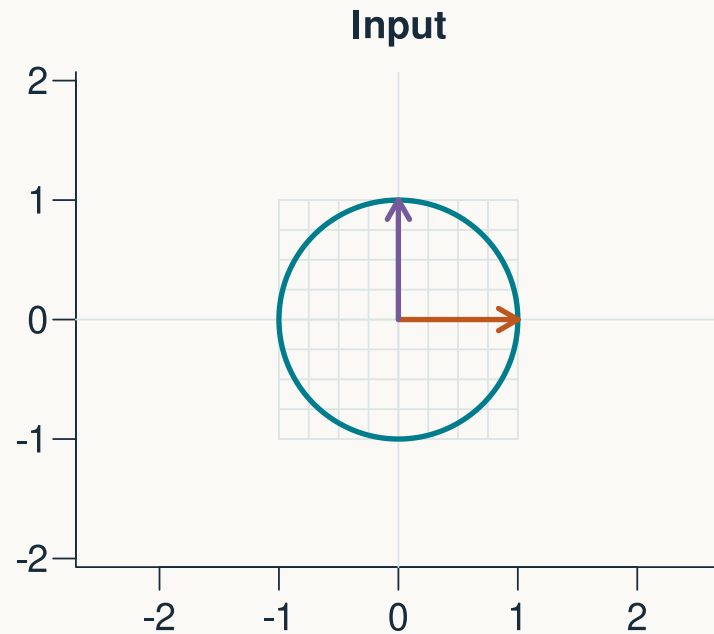
```
1 A %*% B
```

	[,1]	[,2]
[1,]	5	6
[2,]	1	0

Rows meet columns; products are added.

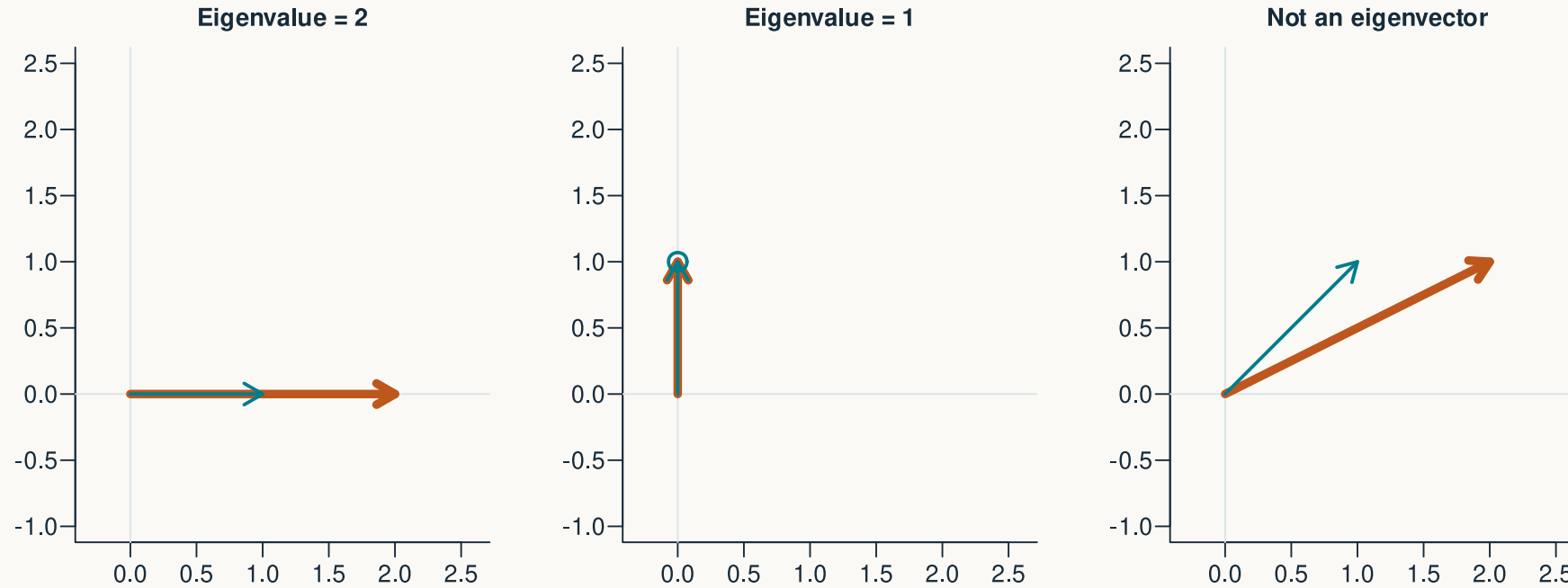
A matrix can transform a whole space

$$\mathbf{S} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{S} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_2 \end{bmatrix}$$



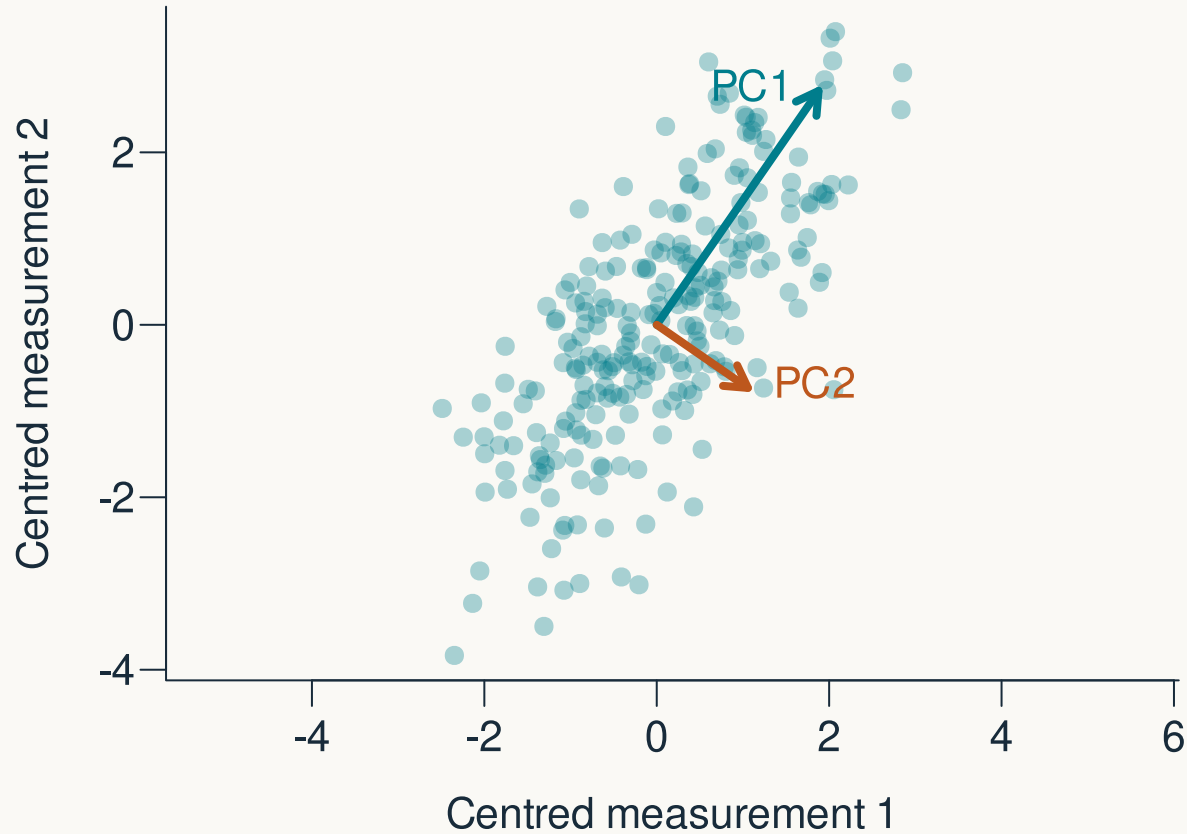
Eigenvectors keep the same line of action

$$S\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}$$



Teal: input. Orange: transformed vector. λ is the eigenvalue.

PCA asks: which direction keeps the most variation?

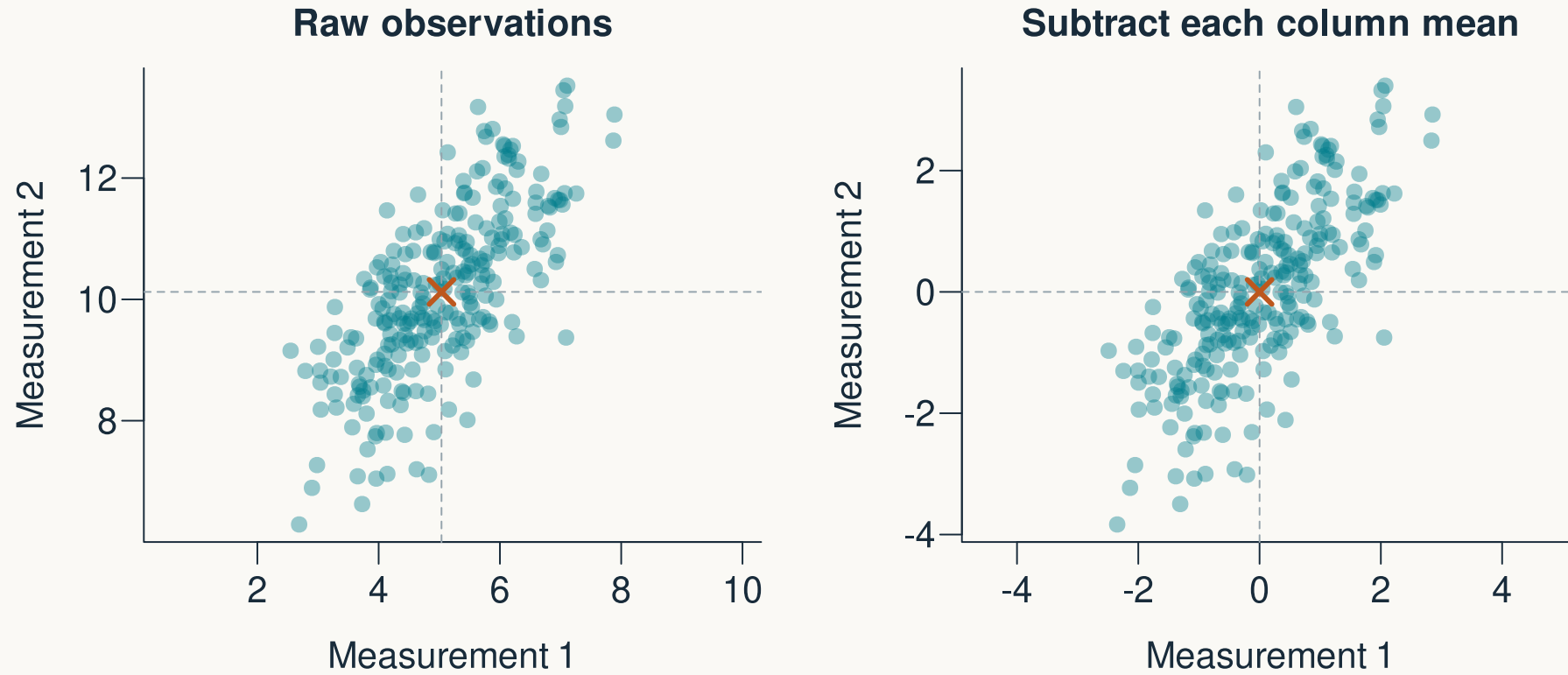


PC1: largest variance of projected scores.

PC2: largest remaining variance, perpendicular to PC1.

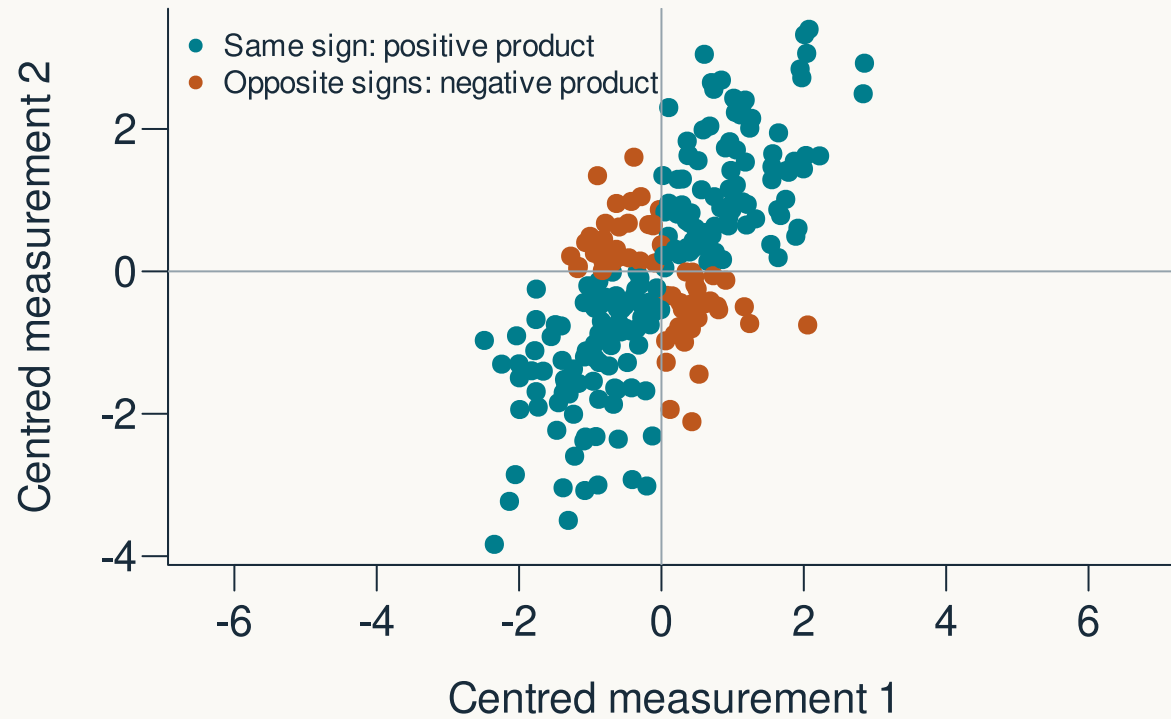
Synthetic data · two related measurements

Step 1: centre each variable



$$X_{c,ij} = X_{ij} - \overline{X_j}$$

Step 2: summarise how centred variables vary together



$$\mathbf{S} = \frac{\mathbf{X}_c^T \mathbf{X}_c}{n - 1}$$

$$\begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_1, x_2) & \text{Var}(x_2) \end{bmatrix}$$

Diagonal: spread.

Off diagonal: variation together.

Step 3: directions and amounts of variation

$$\mathbf{S}\mathbf{v}_j = \lambda_j\mathbf{v}_j$$

PCA object	What it represents
Eigenvector $\mathbf{v}_j$	A unit direction: the loading vector
Eigenvalue λ_j	Variance of scores along that direction
Largest eigenvalue	The first principal component direction

Directions are weights. They are not the observations' coordinates.

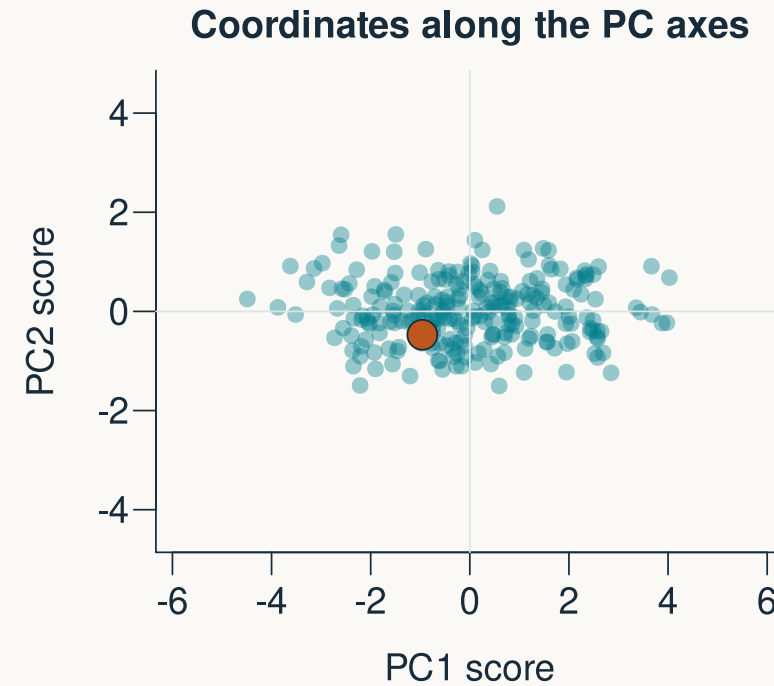
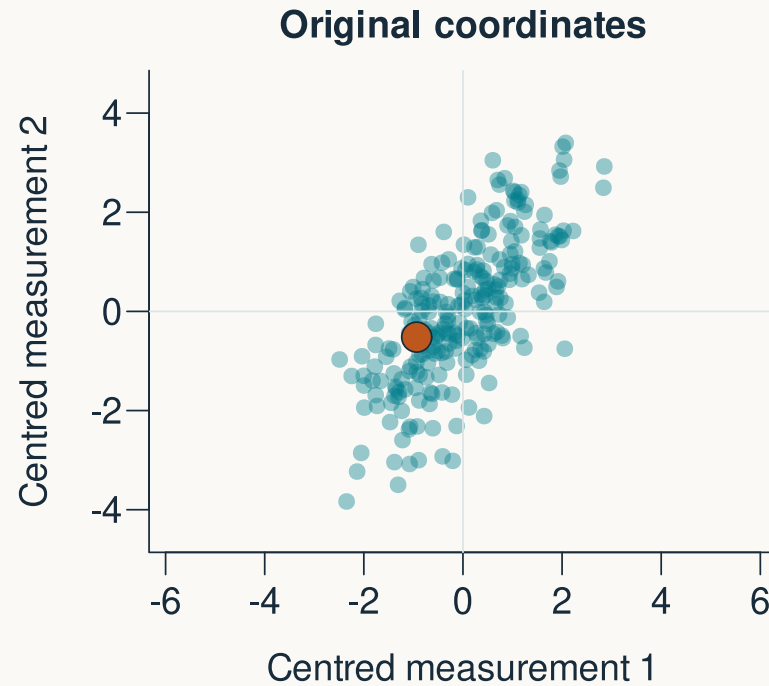
R: build the example and find the directions

```
1  set.seed(2)
2  n <- 250
3  x1 <- rnorm(n)
4  x2 <- x1 + rnorm(n)
5  X <- cbind(x1 + 5, x2 + 10)
6  Xc <- X
7  for (j in 1:ncol(X)) Xc[, j] <- X[, j] - mean(X[, j])
8  S <- t(Xc) %*% Xc / (n - 1)
9  decomposition <- eigen(S, symmetric = TRUE)
10 V <- decomposition$vectors
```

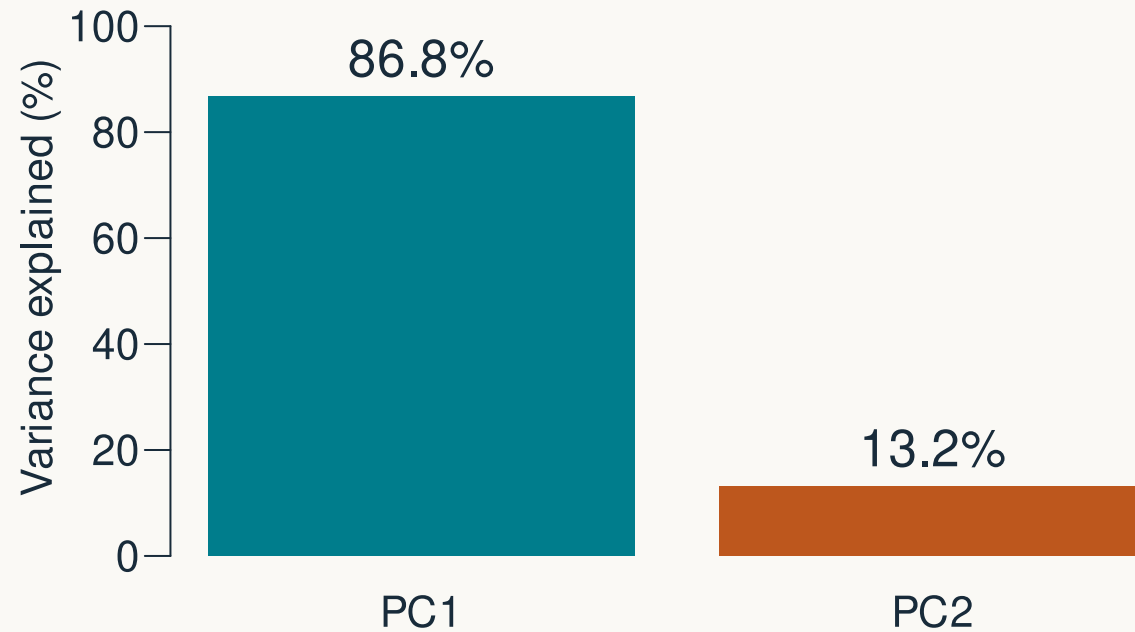
Complete, commented example: [R/pca_example.R](#)

Step 4: scores are the new coordinates

$$\underbrace{\mathbf{Z}}_{n \times p} = \underbrace{\mathbf{X}_c}_{n \times p} \underbrace{\mathbf{V}}_{p \times p}$$



How much variation does each component retain?

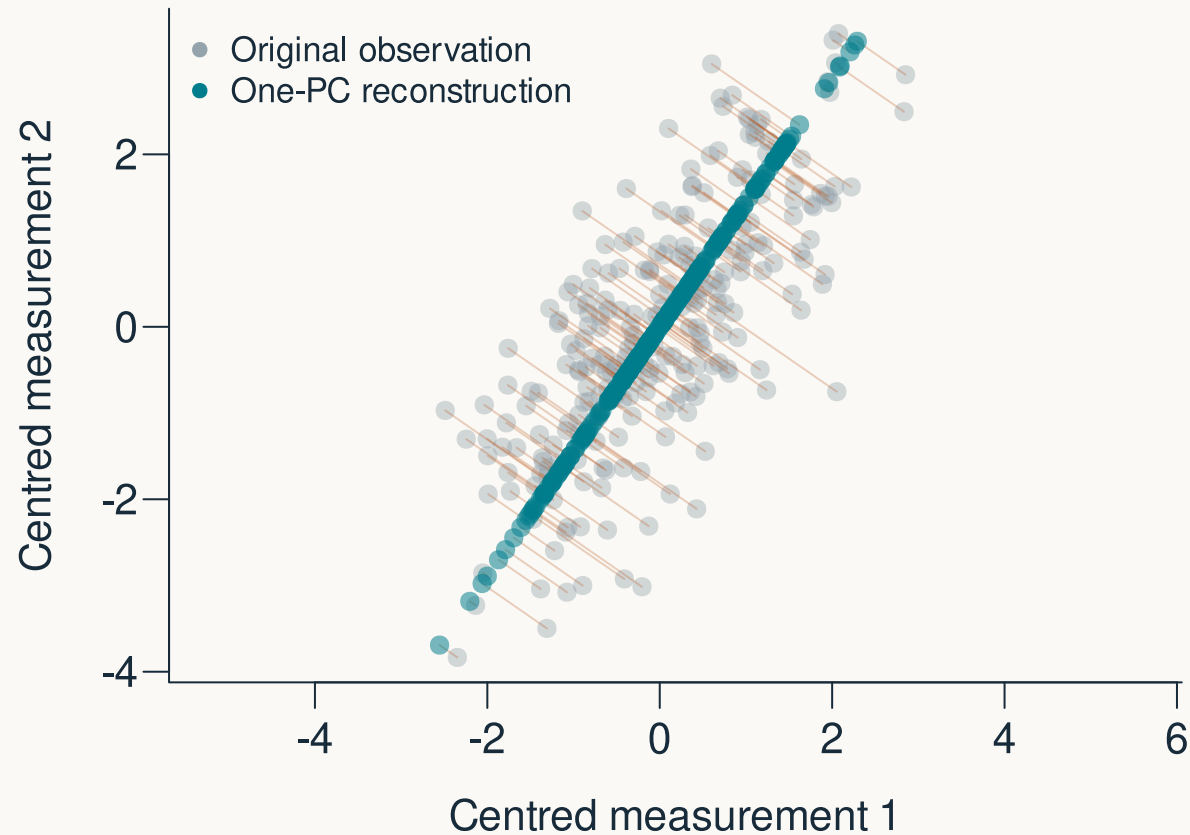


$$\text{fraction for PC } j = \frac{\lambda_j}{\sum_k \lambda_k}$$

PC1 retains **86.8%** here.

High variance does not establish biological importance.

Step 5: keep one score and reconstruct an approximation



$$\hat{\mathbf{x}}_c = z_1 \mathbf{v}_1$$

One coordinate gives a point on a line.

Add the means back to recover the original units and location.

R: project, measure retained variance, reconstruct

```
1 scores <- Xc %*% V
2 lambda <- decomposition$values
3 variance_fraction <- lambda / sum(lambda)
4 z1 <- scores[, 1, drop = FALSE]
5 v1 <- V[, 1, drop = FALSE]
6 Xc_approx <- z1 %*% t(v1)
7 X_approx <- Xc_approx
8 for (j in 1:ncol(X)) {
9   X_approx[, j] <- Xc_approx[, j] + mean(X[, j])
10 }
```



Pause: which changes should affect PCA?

- A.** Add 100 to every value in one column.
- B.** Multiply one column by 100.
- C.** Reverse a loading vector and its corresponding scores.

A: centring removes it. **B:** generally changes covariance PCA.

C: same axis and reconstruction; opposite signs.

In a paper: a trajectory through population activity

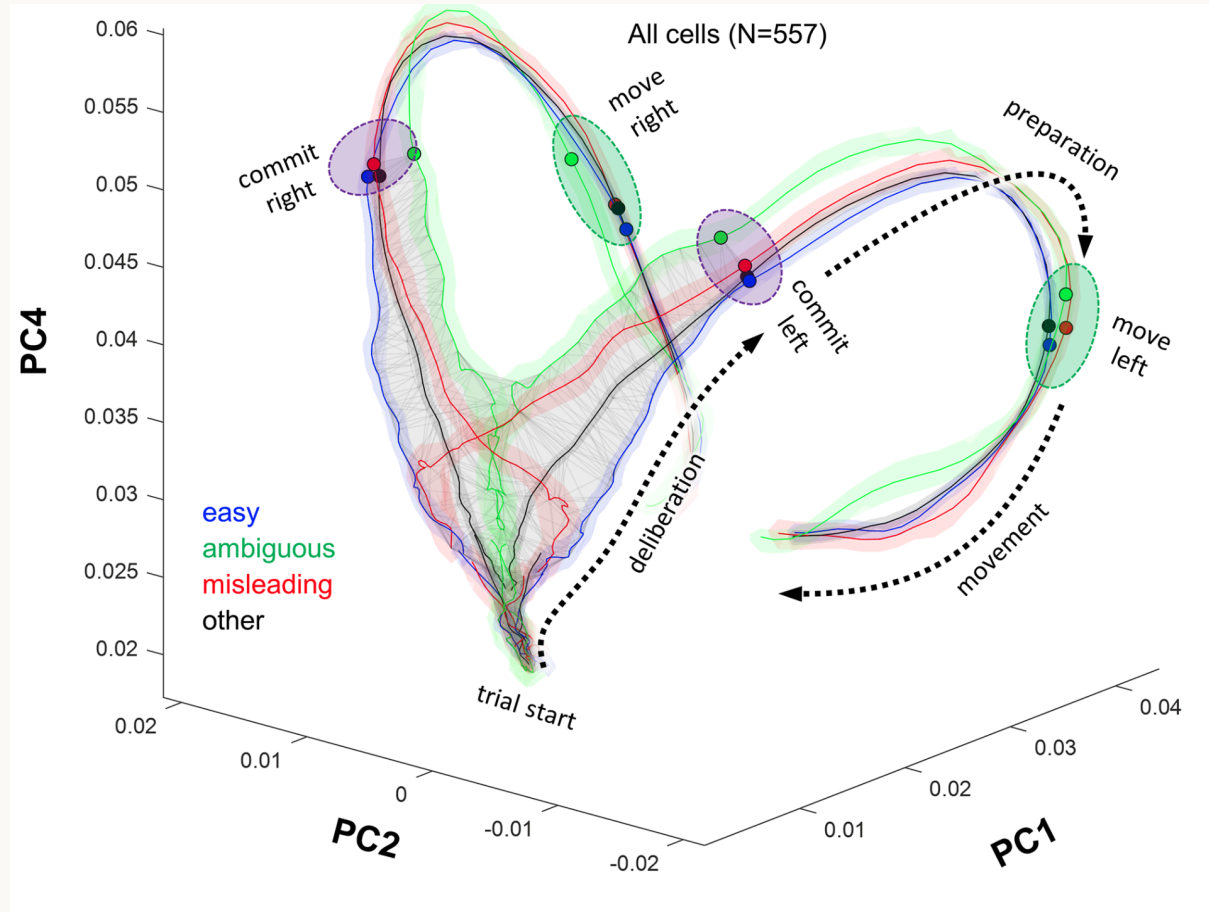


Figure 4A

Each location = population activity at a time point.

PC axes = weighted combinations of neural responses.

The line links successive states.

Thura, Cabana, Feghaly & Cisek (2022), *PLOS Biology*, Figure 4A. [doi:10.1371/journal.pbio.3001861](https://doi.org/10.1371/journal.pbio.3001861). CC BY; existing panel crop from the original teaching assets.

Read the mathematics in a paper

See this...	Read it as...
$\mathbf{Ax}$	A combination of columns; a transformation
$\mathbf{a}^T \mathbf{b}$	Multiply corresponding entries, then add
$\mathbf{Sv} = \lambda \mathbf{v}$	A special direction and its scale factor
$\mathbf{Z} = \mathbf{X}_c \mathbf{V}$	Centred observations in new coordinates

Exit check: how do **loadings**, **scores** and **eigenvalues** differ?

OPTIONAL / Identity and inverse: leave unchanged, or undo

$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{I}\mathbf{x} = \mathbf{x}$$

$$\mathbf{S}^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{S}^{-1}\mathbf{S} = \mathbf{I}$$

An ordinary inverse exists only for a **square, full-rank** matrix.

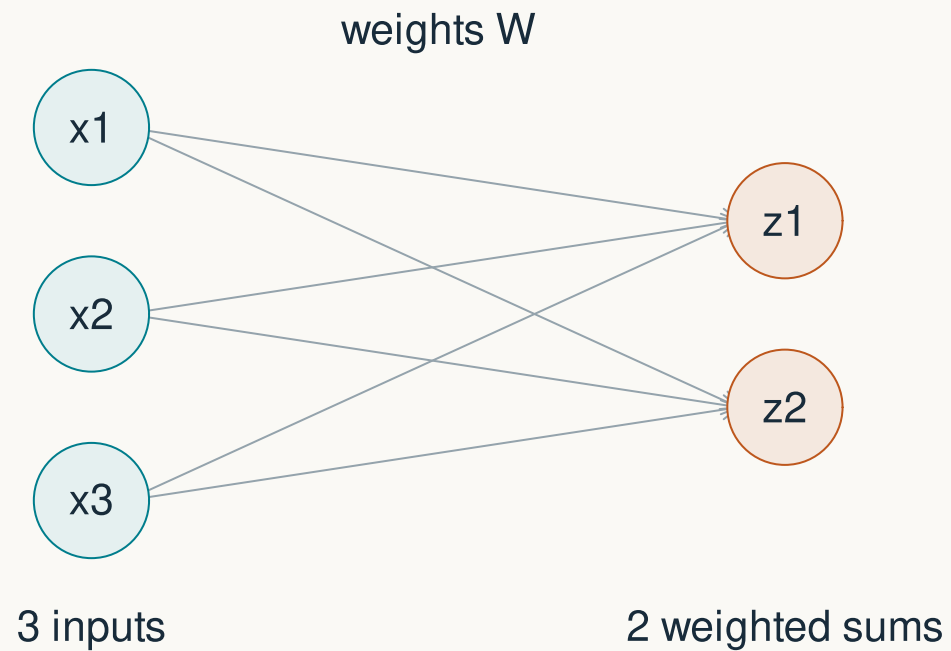
OPTIONAL / A linear model is a matrix-vector product

$$\hat{\mathbf{y}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} 5 \\ 2 \end{bmatrix}}_{\boldsymbol{\beta}} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Rows = observations. Columns = intercept and predictor. Weights = coefficients.

OPTIONAL / A network layer combines inputs using weights

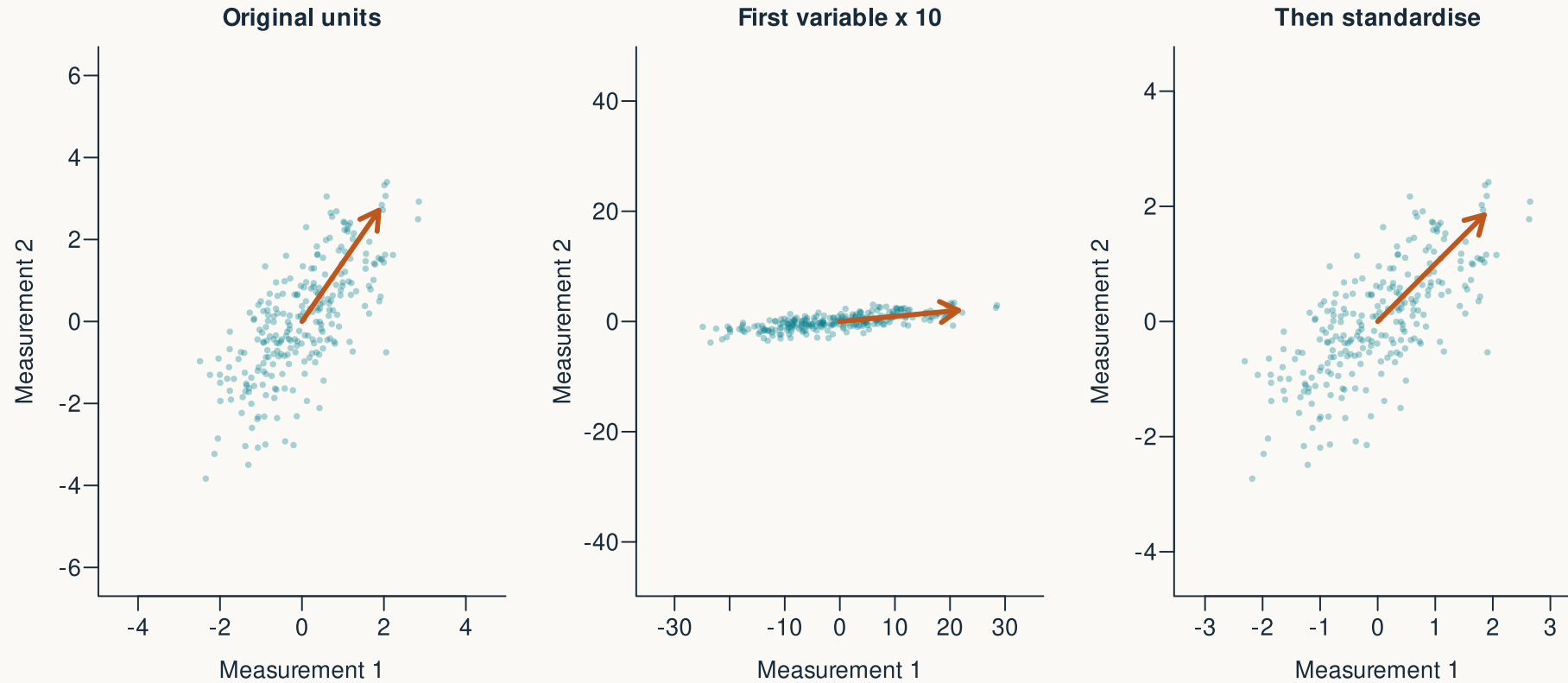


$$\underbrace{\mathbf{z}}_{2 \times 1} = \underbrace{\mathbf{W}}_{2 \times 3} \underbrace{\mathbf{x}}_{3 \times 1} + \underbrace{\mathbf{b}}_{2 \times 1}$$

$$\mathbf{y} = g(\mathbf{z})$$

Weights $\rightarrow$ sums $\rightarrow$ activation.

OPTIONAL / Scaling changes the question PCA answers



Centring removes offsets. **Standardising** also divides by each variable's SD.

OPTIONAL / A few eigenvector caveats

Case	Meaning
$\lambda < 0$	Direction reverses
$\lambda = 0$	The vector maps to zero
$\mathbf{v}$ and $-\mathbf{v}$	The same axis
Repeated eigenvalues	Directions inside that subspace need not be unique

Real symmetric covariance matrices have an orthonormal eigenbasis.
Other real matrices need not have real eigenvectors.